Minor Specialization Project Synopsis

AI-Powered Context-Aware Chatbot for College Information Retrieval

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Abstract

The increasing demand for efficient and real-time access to college-related information has led to the development of intelligent chatbot systems. Traditional systems such as static websites and FAQs suffer from limitations including lack of personalization, poor contextual understanding, and outdated information. This project proposes an advanced AI-powered chatbot for MIT Manipal that leverages Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Natural Language Processing (NLP) to provide accurate, context-aware, and personalized responses. The system integrates institutional APIs, supports multilingual interaction, and incorporates continuous learning mechanisms to improve over time. The chatbot is designed to assist students, faculty, and prospective applicants by delivering dynamic responses related to admissions, courses, placements, campus facilities, and navigation. The proposed system aims to enhance accessibility, improve user engagement, and reduce information retrieval time within academic environments.

Keywords: Chatbot, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Context-Aware Systems, College Information Retrieval

1. Introduction

Access to accurate and up-to-date information is critical in higher education institutions. Students often rely on multiple sources such as websites, brochures, and administrative offices, which can be inefficient and time-consuming. Existing chatbot systems provide basic query handling but lack contextual understanding, personalization, and real-time adaptability.

AI-powered chatbots can significantly improve communication efficiency and user satisfaction in educational environments. This project proposes a next-generation chatbot system for MIT Manipal that addresses these limitations by integrating context awareness, intelligent retrieval systems, and adaptive learning capabilities.

2. Problem Statement

Current college information systems suffer from several critical limitations that hinder effective information access:

- **Lack of context awareness:** Existing systems cannot maintain conversation history or handle multi-turn queries meaningfully.
- **Static and outdated information:** FAQs and static web pages are rarely updated in real time, leading to stale data.
- **Limited personalization:** No differentiation in responses for prospective students, current students, or faculty.
- **Absence of multilingual support:** Most systems are English-only, restricting accessibility.
- **No real-time institutional integration:** Disconnected from live databases, timetables, and admission portals.

These limitations collectively reduce usability and hinder efficient information access across the institution.

3. Objectives

The main objectives of the proposed project are:

1. Develop an AI-powered, context-aware chatbot that maintains conversational memory and delivers coherent multi-turn interactions.
2. Implement Retrieval-Augmented Generation (RAG) for accurate and dynamic information retrieval from college documents and databases.
3. Provide personalized responses tailored to different user types: prospective students, current students, and faculty.
4. Integrate the chatbot with institutional APIs and support multilingual interaction for improved accessibility and real-time data delivery.

4. Proposed System

The proposed system is an AI-powered chatbot that integrates multiple advanced features to address the identified limitations.

4.1 Context-Aware Conversation

The chatbot maintains a conversational memory across sessions, allowing it to understand follow-up queries and provide coherent, multi-turn interactions. This is achieved by storing session-level context and injecting it into every LLM prompt.

4.2 Retrieval-Augmented Generation (RAG)

RAG forms the backbone of the information retrieval pipeline. The process follows these steps:

1. Institutional documents (PDFs, web pages) are preprocessed and chunked.
2. Each chunk is converted into a vector embedding.
3. Embeddings are stored in a vector database (e.g., FAISS or Pinecone).
4. At query time, the most relevant chunks are retrieved and passed as context to the LLM.
5. The LLM generates a grounded, accurate response using the retrieved context.

4.3 Personalization

The chatbot adapts its responses based on the identified user profile:

- **Prospective Students:** Admission procedures, eligibility criteria, fee structures.
- **Current Students:** Timetables, course information, placement updates.
- **Faculty:** Administrative schedules, department resources.

4.4 Multilingual Support

The system supports multiple languages including English, Hindi, and select regional languages, thereby improving accessibility for a diverse user base.

4.5 API Integration

The chatbot is integrated with:

- College databases
- Admission portals
- Timetable and scheduling systems

This enables delivery of real-time, dynamic data to users.

4.6 Continuous Learning System

The chatbot improves over time by:

- Logging user queries and feedback

- Identifying unanswered or poorly answered questions
- Dynamically updating the knowledge base with new information

4.7 Navigation Assistance

The chatbot guides users by providing direct links to relevant web pages and assisting with website navigation, reducing the time users spend searching for resources.

5. System Architecture

The system is composed of the following major components:

Component	Description
User Interface	Web-based chat interface accessible via browser
Backend Server	Handles request routing, session management, and API calls
NLP + LLM Module	Processes natural language queries and generates responses
Vector Database	Stores document embeddings for RAG-based retrieval
API Integration Layer	Connects to institutional databases and external systems
Knowledge Base	Repository of college documents, FAQs, and structured data

System Workflow:

1. User enters a query through the web chat interface.
2. The query is preprocessed and converted into a vector embedding.
3. Relevant document chunks are retrieved from the vector database.
4. The retrieved context and the original query are sent to the LLM.
5. The LLM generates a coherent, grounded response.
6. The response is returned to the user interface in real time.

6. Advantages of the Proposed System

- **Context-aware responses:** Understands follow-up queries and maintains coherent dialogue.
- **Real-time information retrieval:** Always reflects the latest institutional data through live API integration.
- **Improved user experience:** Personalized, fast, and accurate responses across user types.

- **Scalable and adaptable:** Modular architecture allows easy addition of new data sources and features.
- **Reduced administrative load:** Automates repetitive queries, freeing staff for higher-priority tasks.

7. Future Scope

The system can be further enhanced in the following directions:

- **Voice-based interaction:** Integration with speech-to-text and text-to-speech modules.
- **Emotion and sentiment analysis:** Detecting user frustration or confusion to adapt responses accordingly.
- **Predictive recommendation systems:** Proactively suggesting courses, events, or resources based on user history.
- **Mobile application integration:** Extending chatbot access through Android and iOS apps.

8. Conclusion

The proposed AI-powered chatbot aims to revolutionize how students, faculty, and prospective applicants interact with college information systems at MIT Manipal. By incorporating modern AI techniques such as Retrieval-Augmented Generation, context-aware conversation management, user personalization, and multilingual support, the system offers a scalable, efficient, and user-friendly solution for academic institutions. This project represents a significant step towards intelligent, real-time academic assistance systems that can meaningfully reduce information retrieval friction in higher education.